

Ayan Khan

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Portfolio | GitHub | LinkedIn | LeetCode

Profile

Bachelor of Computer Applications student specializing in Machine Learning and Data Science, with a strong foundation in full-stack development, object-oriented programming, algorithms, and computational problem solving. Experienced in building applications using the MERN stack and developing modular C++ projects involving stochastic modeling and compiler design. Interested in building reliable software and contributing to real-world engineering projects.

Education

Bachelor of Computer Applications – Machine Learning and Data Science 2025 – 2028

Deen Dayal Upadhyaya Gorakhpur University, India

SGPA: 9.08/10.00

Relevant Study: Data Structures and Algorithms, Object-Oriented Programming, Statistics, Exploratory Data Analysis, Databases, Machine Learning Fundamentals

Technical Skills

Languages: JavaScript, C++, Python, C, Java

Frontend: React.js, Vite, Tailwind CSS, React Router, Axios, Responsive Web Design

Backend and Database: Node.js, Express.js, REST APIs, MongoDB, Mongoose, JWT Authentication, bcrypt.js

Core: Data Structures and Algorithms, Object-Oriented Programming, Complexity Analysis, Modular Design, Debugging, Testing, Error Handling

Tools: Git, GitHub, Linux, VS Code, Command Line, Postman, Jupyter Notebook, C++ STL

Projects

Employee Management System

MongoDB, Express.js, React.js, Node.js

- Built a full-stack employee management application with JWT authentication, protected routes, and complete employee CRUD operations.
- Developed RESTful APIs using Express and MongoDB with Mongoose schemas, search and department filtering, validation, and dashboard statistics.
- Created a responsive React interface using Vite and Tailwind CSS, with Axios-based API integration, routing, forms, and user notifications.

Markov Text Generator

C++, Markov Chains, CLI

- Built a modular text generator using fixed-order Markov chains, probabilistic token sampling, and state-transition modeling.
- Implemented tokenization, model training, transition-table construction, and generation as independently testable components.

Prometheus Compiler

C++, Compiler Design

- Developing an experimental compiler with lexical analysis, tokenization, parsing, and Abstract Syntax Tree generation.
- Applied modular architecture, object-oriented programming, error handling, debugging, testing, and Git-based version control.

Technical Activities and Achievements

- Solved 223+ LeetCode problems and achieved a 1500+ contest rating, strengthening algorithm design, complexity analysis, and debugging skills.
- Maintained a repository of 200+ C++ and Python solutions with documented approaches and time-space complexity analysis.
- Completed the First Contributions GitHub workflow using forks, branches, commits, pull requests, and code review.